

Computational Methods #2

Problem 1

For the new special function I chose the Hermite polynomials $H_n(x)$. Unlike the Legendre polynomials they do not have a range $[-1, 1]$. Some code had to be added to normalize the output to fit on the generated canvas.

After the array values[] is populated I find the largest value in the array and divide by it to set the largest value to 1 before plotting.

Problem 2

By comparing the output of the function of $\gamma = 1/\sqrt{1-\beta^2}$ and $\gamma = 1/\sqrt{(2-\epsilon)\epsilon}$ we find that at $\epsilon = 1.0 \times 10^{-13}$ we get a relative difference of $\Delta\gamma/\gamma = 0.000155436$. This corresponds to a β of $\beta_{\max} = 1.0 - 1.0 \times 10^{-13}$.

Problem 3

The solution of the time independent Schrodinger equation is

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0 \\ te^{inkx}, & x > 0 \end{cases}$$

Where n is the “refractive index” of the second medium given by $n = \sqrt{1-V/E}$. The key to solving for $\psi_{k,n}(x)$ is finding r and t . By using continuity of the wave function $\psi(0^+) = \psi(0^-)$ and the differentiability of the wave function $\psi'(0^+) = \psi'(0^-)$ we get the following set of equations for r and t .

$$\begin{aligned} 1 + r &= t \\ -1 + r &= nt \end{aligned}$$

Solving these equations for r and t we get

$$r = \frac{1-n}{1+n}, \quad t = \frac{2}{1+n}$$

Using the probability current we can find the transmission and reflection coefficients.

$$\begin{aligned} j^T &= \frac{\hbar}{m} \text{Im} [\psi_T d\psi_T^*/dx] \\ &= \frac{\hbar k}{m} \text{Re} [n |t|^2] \end{aligned}$$

The transmission coefficient $T = \text{Re}[n] |t|^2$. By continuity we know that $R = 1 - T$. This implies that when $E < V$ the transmission probability vanishes. This is because this is not a potential step, so the transmitted wave decays to zero instead of decaying until it reaches the second barrier.

Problem 4

A set of input angles were tested and confirmed that the output of $\sinh(x)$ and $\cosh(x)$ matches the output of $\sin(x)$ and $\cos(x)$.